

Chapter 9, Problem 6PP

Problem

If $v_1 = -10 \sin(\omega t - 30^\circ)$ V and $v_2 = 20 \cos(\omega t + 45^\circ)$ V, find $v = v_1 + v_2$.

Step-by-step solution

Step 1 of 2

Find the signal $v = v_1 + v_2$.

Write the expression for v in phasor form.

$$\mathbf{V} = \mathbf{V}_1 + \mathbf{V}_2$$

Substitute $10\angle 60^\circ$ V for $\mathbf{V}_1$, $20\angle 45^\circ$ V for $\mathbf{V}_2$ in this expression.

$$\begin{aligned}\mathbf{V} &= 10\angle 60^\circ \text{ V} + 20\angle 45^\circ \text{ V} \\ &= 5 + j8.66 + 14.14 + j14.14 \\ &= 19.14 + j22.8 \\ &= 29.7\angle 49.98^\circ\end{aligned}$$

Therefore, the expression for v in phasor form is $29.7\angle 49.98^\circ$ V.

Step 2 of 2

Find the sinusoid corresponding to the phasor $\mathbf{V} = 29.7\angle 49.98^\circ$ V.

Write the expression for the sinusoid corresponding to the phasor $\mathbf{V} = 29.7\angle 49.98^\circ$ V.

$$\begin{aligned}\mathbf{V} &= 29.7\angle 49.98^\circ \text{ V} \\ &= 29.77 \cos(\omega t + 49.987^\circ) \text{ V}\end{aligned}$$

Therefore, the required expression for $v(t)$ is $\boxed{29.77 \cos(\omega t + 49.987^\circ) \text{ V}}$.